NAME: _____ Geometry (FL B.E.S.T.)

4.5 Hypotenuse-Leg (HL) Congruence

The **HL Theorem** is a congruence shortcut that applies only to right triangles: if the hypotenuse and one leg of a right triangle are congruent to the hypotenuse and corresponding leg of another right triangle, the triangles are congruent. HL is really a special case of "SSA," which is normally not a valid shortcut — but for right triangles it works because the Pythagorean Theorem *uniquely* determines the missing leg once the hypotenuse and one leg are fixed ($b = \sqrt{c^2 - a^2}$ has only one positive solution). That forces all three sides to match, turning the situation into SSS. This worksheet applies HL numerically, algebraically, and with coordinates, and contrasts it with SAS when two legs (rather than the hypotenuse) are given.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. HL is a valid shortcut for which kind of triangles?

2. HL requires the hypotenuse and what else to be congruent?

3. What is the name of the side opposite the right angle?

4. Right angle at C in $\triangle ABC$. Which side is the hypotenuse?

5. Two right triangles have a congruent hypotenuse and a congruent leg. Which postulate applies?

6. A shared leg between two right triangles is congruent to itself by which property?

7. Two legs (not the hypotenuse) and the right angle between them are congruent. Which postulate applies?

8. Can HL be used to prove two acute (non-right) triangles congruent?

9. After proving triangles congruent by HL, corresponding parts are congruent by what idea?

10. In a right triangle, the hypotenuse is always the ____ side.

11. Right triangles with hypotenuses $\overline{AB} \cong \overline{DE}$ and legs $\overline{BC} \cong \overline{EF}$. Which postulate applies?

12. Must both triangles be confirmed as right triangles before applying HL?

13. Two right triangles have hypotenuses $6x - 9$ and 39, and a corresponding leg of 15 in each. Find x, then find the other leg using the Pythagorean Theorem.

14. Isosceles $\triangle ABC$ has $AB = AC = 17$ and base $BC = 16$. Altitude $\overline{AD}$ meets $\overline{BC}$ at D. Using HL/CPCTC (the altitude bisects the base), find BD, DC, and AD.

15. Right $\triangle ABC$ (right angle C) has hypotenuse $AB = 25$, leg $BC = 7$. Right $\triangle DEF$ (right angle F) has hypotenuse $DE = 25$, leg $EF = 24$. Determine whether the triangles must be congruent, and state a valid correspondence.

✓ Answer Key

1. Right triangles
2. One leg
3. Hypotenuse
4. $\overline{AB}$
5. HL
6. Reflexive Property
7. SAS
8. No
9. CPCTC
10. Longest
11. HL
12. Yes
13. $x = 8$; hypotenuse $= 39$; other leg $= 36$
14. $BD = DC = 8$; $AD = 15$
15. Yes — both are 7-24-25 triangles ($AC = 24$; $\triangle DEF$'s missing leg $= 7$); congruent by HL with $C \leftrightarrow F$, $B \leftrightarrow E$ (leg 7), $A \leftrightarrow D$ (leg 24)